

From: [Erriques Francesco](#)
To: [Erriques Francesco](#); francesco.erriques@gmail.com; british_language@libero.it
Subject: basis vector derivatives
Date: lunedì 15 giugno 2026 16:53:51
Attachments: [image001.png](#)
[image002.png](#)
[image003.png](#)
[image004.png](#)
[image005.png](#)
[image006.png](#)
[image007.png](#)
[image008.png](#)
[image009.png](#)
[image010.png](#)
[image011.png](#)
[image012.png](#)
[image013.png](#)
[image014.png](#)
[image015.png](#)
[image016.png](#)
[image017.png](#)
[image018.png](#)
[image019.png](#)
[image020.png](#)
[image021.png](#)
[image022.png](#)
[image023.png](#)
[image024.png](#)
[image025.png](#)
[image026.png](#)
[image027.png](#)
[image028.png](#)
[image029.png](#)
[image030.png](#)
[image031.png](#)
[image032.png](#)
[image033.png](#)
[image034.png](#)
[image035.png](#)
[image036.png](#)
[image037.png](#)

Great question—this is actually the most geometric way to understand Christoffel symbols.

We'll now derive them using **derivatives of the coordinate basis vectors** in polar coordinates and show how this matches the metric-based result.

1. Position vector and basis vectors

In the plane, the position vector is:

$$\mathbf{x}(r, \theta) = r \hat{\mathbf{e}}_r$$

where the **polar unit vectors** are:

$$\hat{\mathbf{e}}_r = (\cos \theta, \sin \theta), \hat{\mathbf{e}}_\theta = (-\sin \theta, \cos \theta)$$

Coordinate basis

The coordinate basis vectors are:

$$\mathbf{e}_r = \frac{\partial \mathbf{x}}{\partial r} = \hat{\mathbf{e}}_r$$

$$\mathbf{e}_\theta = \frac{\partial \mathbf{x}}{\partial \theta} = r \hat{\mathbf{e}}_\theta$$

So:

$$\boxed{\mathbf{e}_r = \hat{\mathbf{e}}_r, \mathbf{e}_\theta = r \hat{\mathbf{e}}_\theta}$$

Note: this is a **non-orthonormal coordinate basis**, because:

$$|\mathbf{e}_\theta| = r$$

2. Key derivatives of unit vectors

We use the standard identities:

$$\frac{\partial \hat{\mathbf{e}}_r}{\partial r} = 0, \frac{\partial \hat{\mathbf{e}}_r}{\partial \theta} = \hat{\mathbf{e}}_\theta$$
$$\frac{\partial \hat{\mathbf{e}}_\theta}{\partial r} = 0, \frac{\partial \hat{\mathbf{e}}_\theta}{\partial \theta} = -\hat{\mathbf{e}}_r$$

3. Compute derivatives of the basis vectors

The Christoffel symbols are defined through:

$$\frac{\partial \mathbf{e}_i}{\partial x^j} = \Gamma_{ij}^k \mathbf{e}_k$$

We now compute all these derivatives.

(a) Derivatives of $\mathbf{e}_r$

With respect to r :

$$\frac{\partial \mathbf{e}_r}{\partial r} = \frac{\partial \hat{\mathbf{e}}_r}{\partial r} = 0$$

So:

$$\Gamma_{rr}^k = 0$$

With respect to θ :

$$\frac{\partial \mathbf{e}_r}{\partial \theta} = \hat{\mathbf{e}}_\theta$$

But we must express this in the **coordinate basis**:

$$\hat{\mathbf{e}}_\theta = \frac{1}{r} \mathbf{e}_\theta$$

Thus:

$$\frac{\partial \mathbf{e}_r}{\partial \theta} = \frac{1}{r} \mathbf{e}_\theta$$

So:

$$\Gamma_{r\theta}^\theta = \frac{1}{r}, \Gamma_{r\theta}^r = 0$$

(b) Derivatives of $\mathbf{e}_\theta$

Recall:

$$\mathbf{e}_\theta = r \hat{\mathbf{e}}_\theta$$

With respect to r :

$$\frac{\partial \mathbf{e}_\theta}{\partial r} = \hat{\mathbf{e}}_\theta$$

Rewrite in basis:

$$\hat{\mathbf{e}}_\theta = \frac{1}{r} \mathbf{e}_\theta$$

Thus:

$$\frac{\partial \mathbf{e}_\theta}{\partial r} = \frac{1}{r} \mathbf{e}_\theta$$

So:

$$\Gamma_{\theta r}^\theta = \frac{1}{r}$$

With respect to θ :

$$\frac{\partial \mathbf{e}_\theta}{\partial \theta} = \frac{\partial}{\partial \theta} (r \hat{\mathbf{e}}_\theta) = r \frac{\partial \hat{\mathbf{e}}_\theta}{\partial \theta} = -r \hat{\mathbf{e}}_r$$

But:

$$\hat{\mathbf{e}}_r = \mathbf{e}_r$$

So:

$$\frac{\partial \mathbf{e}_\theta}{\partial \theta} = -r \mathbf{e}_r$$

Thus:

$$\Gamma_{\theta\theta}^r = -r$$

4. Final result

From basis derivatives, we get exactly:

$$\Gamma_{\theta\theta}^r = -r, \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$$

All others vanish.

5. Connection with the metric (deep insight)

This derivation clarifies something very important:

- The Christoffel symbols encode how the **basis vectors change in space**
- In polar coordinates:
 - $\mathbf{e}_r$ rotates as θ changes
 - $\mathbf{e}_\theta$ changes both direction and magnitude

Meanwhile:

- The metric:

$$ds^2 = dr^2 + r^2 d\theta^2$$
 already “knows” this variation

And in fact:

$$\text{metric derivatives} \Leftrightarrow \text{basis vector variation}$$

These are two equivalent ways to get the same connection coefficients.

6. Geometric interpretation

- $\Gamma_{r\theta}^\theta = 1/r$: $\rightarrow$ angular basis vector changes when radius changes
- $\Gamma_{\theta\theta}^r = -r$: $\rightarrow$ moving along θ produces inward radial change

These are exactly what you expect from circular motion.

☑ Summary

- Starting from:

$$\mathbf{e}_r = \hat{\mathbf{e}}_r, \mathbf{e}_\theta = r \hat{\mathbf{e}}_\theta$$
- Differentiating and expressing back into the same basis yields:

$$\Gamma_{ij}^k$$
- These match perfectly the ones derived from the metric tensor.

If you want, the next natural step is to derive the **geodesic equations in polar coordinates** and see how straight lines in Cartesian coordinates emerge from these Christoffel symbols.

FRANCESCO ERRIQUES

Informationstechnologie
Tecnologia dell'informazione

Mobil: +39 331.63.42.823
E-Mail: francesco.erriques@bbt-se.com
Web: www.bbt-se.com

GALLERIA DI BASE DEL BRENNERO - BRENNER BASISTUNNEL BBT SE

Piazza Stazione 1 • IT-39100 Bolzano
Tel.: +39 0471.062.273 • Fax: +39 0471.062.211